# Automata,complexity and computability Theory. Comprehensive Question Bank

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# Part 1: Finite Automata & Conversions

## QUESTION ONE

Convert the following Non-Deterministic Finite Automaton (NFA) into its equivalent Deterministic Finite Automaton (DFA).

Table 1: NFA Transition Table for Question One

| State | Input 0 | Input 1 |
|---|---|---|
| $\to p$ | $\{p, q\}$ | $\{p\}$ |
| $q$ | $\{r\}$ | $\{r\}$ |
| $r$ | $\{s\}$ | $\emptyset$ |
| $*s$ | $\{s\}$ | $\{s\}$ |

## QUESTION TWO

Convert the following NFA into its equivalent DFA.

Table 2: NFA Transition Table for Question Two

| State | Input 0 | Input 1 |
|---|---|---|
| $\to p$ | $\{q, s\}$ | $\{q\}$ |
| $*q$ | $\{r\}$ | $\{q, r\}$ |
| $r$ | $\{s\}$ | $\{p\}$ |
| $*s$ | $\emptyset\bullet$ | $\{p\}$ |

## QUESTION THREE

Consider the following $\epsilon$-NFA configuration:

Table 3: $\epsilon$-NFA Transition Table for Question Three

| State | $\epsilon$ | Input a | Input b | Input c |
|---|---|---|---|---|
| $\to p$ | $\emptyset$ | $\{p\}$ | $\{q\}$ | $\{r\}$ |
| $q$ | $\{p\}$ | $\{q\}$ | $\{r\}$ | $\emptyset$ |
| $*r$ | $\{q\}$ | $\{r\}$ | $\emptyset$ | $\{p\}$ |

(a) Compute the $\epsilon$-closure of each state.

(b) Give all the strings of length three or less accepted by the automaton.

(c) Convert the automaton to a DFA.

### QUESTION FOUR

Repeat Question Three completely for the following structural $\epsilon$-NFA:

Table 4: $\epsilon$-NFA Transition Table for Question Four

| **State** | $\epsilon$ | **Input a** | **Input b** | **Input c** |
|---|---|---|---|---|
| $\rightarrow p$ | $\{q, r\}$ | $\{q\}$ | $\{q\}$ | $\{r\}$ |
| $q$ | $\emptyset$ | $\{p\}$ | $\{r\}$ | $\{p, q\}$ |
| $*r$ | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ |

## Part 2: Regular Expressions & Language Design

### QUESTION FIVE

Write regular expressions for the following languages:

(a) The set of strings over alphabet $\{a, b, c\}$ containing at least one $a$ and at least one $b$.

(b) The set of strings of 0's and 1's whose tenth symbol from the right end is 1.

(c) The set of strings of 0's and 1's with at most one pair of consecutive 1's.

### QUESTION SIX

Write regular expressions for the following languages:

(a) The set of all strings of 0's and 1's such that every pair of adjacent 0's appears before any pair of adjacent 1's.

(b) The set of strings of 0's and 1's whose number of 0's is divisible by five.

### QUESTION SEVEN

Write regular expressions for the following languages:

(a) The set of all strings of 0's and 1's not containing 101 as a substring.

(b) The set of all strings with an equal number of 0's and 1's, such that no prefix has two more 0's than 1's, nor two more 1's than 0's.

(c) The set of strings of 0's and 1's whose number of 0's is divisible by five and whose number of 1's is even.

### QUESTION EIGHT

Give English descriptions of the languages of the following regular expressions:

(a) $(1 + \epsilon)(001)^*0$

(b) $(01^*)000(0 + 1)^*$

(c) $(0 + 10)^*1$

### QUESTION NINE

In Example 3.1 we pointed out that $\emptyset$ is one of two languages whose closure is finite. What is the other?

## Part 3: State Elimination & Algebraic Laws

### QUESTION TEN

Here is a transition table for a DFA:

Table 5: DFA Transition Table for Question Ten

| State | Input 0 | Input 1 |
|---|---|---|
| $\rightarrow q_1$ | $q_2$ | $q_1$ |
| $q_2$ | $q_3$ | $q_1$ |
| $*q_3$ | $q_3$ | $q_2$ |

(a) Give all the regular expressions $R_{ij}^{(0)}$. *Note: Think of state $q_i$ as if it were the state with integer number $i$.*

(b) Give all the regular expressions $R_{ij}^{(1)}$. Try to simplify the expressions as much as possible.

(c) Give all the regular expressions $R_{ij}^{(2)}$. Try to simplify the expressions as much as possible.

(d) Give a regular expression for the language of the automaton.

(e) Construct the transition diagram for the DFA and give a regular expression for its language by eliminating state $q_2$.

### QUESTION ELEVEN

Convert the following DFA to a regular expression, using the state-elimination technique of Section 3.2.2.

Table 6: DFA Transition Table for Question Eleven

| State | Input 0 | Input 1 |
|---|---|---|
| $\rightarrow *p$ | $s$ | $p$ |
| $q$ | $p$ | $s$ |
| $r$ | $r$ | $q$ |
| $s$ | $q$ | $r$ |

### QUESTION TWELVE

Convert the following regular expressions to NFA's with $\epsilon$-transitions.

(a) $01^*$

(b) $(0+1)01$

(c) $00(0+1)^*$

### QUESTION THIRTEEN

Eliminate $\epsilon$-transitions from your $\epsilon$-NFA's of Question Twelve.

## QUESTION FOURTEEN

Let $A = (Q, \Sigma, \delta, q_0, \{q_f\})$ be an $\epsilon$-NFA such that there are no transitions into $q_0$ and no transitions out of $q_f$. Describe the language accepted by each of the following modifications of $A$, in terms of $L = L(A)$:

(a) The automaton constructed from $A$ by adding an $\epsilon$-transition from $q_f$ to $q_0$.

(b) The automaton constructed from $A$ by adding an $\epsilon$-transition from $q_0$ to every state reachable from $q_0$.

(c) The automaton constructed from $A$ by adding an $\epsilon$-transition to $q_f$ from every state that can reach $q_f$ along some path.

(d) The automaton constructed from $A$ by doing both (b) and (c).

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## Part 4: The Pumping Lemma & Non-Regular Languages

### QUESTION FIFTEEN

At the beginning of Section 3.4.6, we gave part of a proof that $(L+M)^* = (L^*M^*)^*$. Complete the proof by showing that strings in $(LM^*)^*$ are also in $(L+M)^*$.

### QUESTION SIXTEEN

Complete the proof of Theorem 3.13 by handling the cases where regular expression $E$ is of the form $FG$ or of the form $F^*$.

### QUESTION SEVENTEEN

Prove that the following are not regular languages:

(a) $\{0^n1^n \mid n \geq 1\}$. Apply the pumping lemma directly to your proof.

(b) The set of strings of balanced parentheses.

(c) $\{0^n10^n \mid n \geq 1\}$.

(d) $\{0^n1^m2^n \mid n \text{ and } m \text{ are arbitrary integers}\}$.

(e) $\{0^n1^m \mid n \leq m\}$.

(f) $\{0^n1^{2n} \mid n \geq 1\}$.

### QUESTION EIGHTEEN

Prove that the following are not regular languages:

(a) $\{0^n \mid n \text{ is a perfect square}\}$.

(b) $\{0^n \mid n \text{ is a perfect cube}\}$.

(c) $\{0^n \mid n \text{ is a power of } 2\}$.

(d) The set of strings of 0's and 1's whose length is a perfect square.

(e) The set of strings of 0's and 1's that are of the form $ww$, that is, some string repeated.

(f) The set of strings of 0's and 1's that are of the form $ww^R$, that is, some string followed by its reverse.

(g) The set of strings of 0's and 1's of the form $w\overline{w}$, where $\overline{w}$ is formed from $w$ by replacing all 0's by 1's, and vice-versa.

(h) The set of strings of the form $w1^n$, where $w$ is a string of 0's and 1's of length $n$.

### QUESTION NINETEEN

When we try to apply the pumping lemma to a regular language, the "adversary wins," and we cannot complete the proof. Show what goes wrong when we choose $L$ to be one of the following languages:

(a) The empty set.

(b) $\{00, 11\}$.

(c) $(00 + 11)^*$.

(d) $010^*1$.

# Part 5: Homomorphisms & Closure Properties

## QUESTION TWENTY

Suppose $h$ is the homomorphism from the alphabet $\{0, 1, 2\}$ to the alphabet $\{a, b\}$ defined by: $h(0) = a$; $h(1) = ab$, and $h(2) = ba$.

(a) What is $h(0120)$?

(b) What is $h(21120)$?

(c) If $L$ is the language $L(01^*2)$, what is $h(L)$?

(d) If $L$ is the language $L(0 + 12)$, what is $h(L)$?

(e) Suppose $L$ is the language $\{ababa\}$, determine $h^{-1}(L)$.

## QUESTION TWENTY-ONE

Show that the regular languages are closed under the following operations:

(a) $\min(L) = \{w \mid w \text{ is in } L, \text{ but no proper prefix of } w \text{ is in } L\}$.

(b) $\max(L) = \{w \mid w \text{ is in } L \text{ and for no } x \text{ other than } \epsilon \text{ is } wx \text{ in } L\}$.

(c) $\text{init}(L) = \{w \mid \text{for some } x, wx \text{ is in } L\}$.

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## Part 6: Comprehensive Applied Exam Problems

### QUESTION TWENTY-TWO

Suppose you are the Artificial Intelligence consultant for a popular software company that created an application to provide useful customer statistics to restaurants and supermarkets. The architectural $\epsilon$-NFA below was modeled to parse the intelligent classification matrix inputs:

Table 7: $\epsilon$-NFA Transition Rules for Question Twenty-Two

| State | $\epsilon$ | Input a | Input b | Input c |
|---|---|---|---|---|
| $\rightarrow 1$ | $\emptyset$ | $\{1\}$ | $\{2\}$ | $\{3\}$ |
| 2 | $\{1\}$ | $\{2\}$ | $\{3\}$ | $\emptyset$ |
| $*3$ | $\{2\}$ | $\{3\}$ | $\emptyset$ | $\{1\}$ |

(a) Convert the $\epsilon$-NFA above to an equivalent DFA (10 marks).

(b) Convert the automaton above to its clean regular expressions (10 marks).

(c) Minimize the DFA to come up with the minimal structural engine using (16 marks):

  i) The set partition or double complement method.
  ii) The unreachable states method if any.
  iii) Method for handling structural components with several final states.

### QUESTION TWENTY-THREE •

A transportation analytics problem requires you to write a program function that determines the number of the latest models of cars that pass through an intersection in a single day. This function counts the total number of cars that pass every hour, from which it has to deduct the old models known by the system, thus yielding the true count of the latest models. Design and write down the formal transitions of a **Turing Machine** to model this essential subtraction operation. (15 Marks)

### QUESTION TWENTY-FOUR

Intrusion detection systems are deployed to search for structural adversarial patterns within high-throughput file streams. These tracking elements are defined by DFA frameworks. Perform state minimization optimizations on both architectural systems below:

**(i) DFA Configuration One**

Table 8: DFA Configuration One Transition Table

| State | Input a | Input b |
|---|---|---|
| $\rightarrow 1$ | 2 | 4 |
| 2 | 2 | 3 |
| 3 | 3 | 2 |
| 4 | 1 | 4 |

### (ii) DFA Configuration Two

Table 9: DFA Configuration Two Transition Table

| State | Input 0 | Input 1 |
|---|---|---|
| $\rightarrow q_0$ | $q_1$ | $q_0$ |
| $q_1$ | $q_1$ | $q_2$ |
| $*q_2$ | $q_2$ | $q_2$ |

(a) Convert the following DFA systems into regular expressions (12 marks).

(b) Minimize the DFA's below to come up with minimal structural configurations (16 marks):

  a. Using the set partition or double complement method.
  b. Using the unreachable states method if any.
  c. Using the processing method for environments having several final states.

## QUESTION TWENTY-FIVE

Consider the 8-state configuration tracking automaton $N$ below:

Table 10: Automaton $N$ Transition Matrix Profile

| State | Input 0 | Input 1 |
|---|---|---|
| $\rightarrow A$ | $B\bullet$ | $A$ |
| $B$ | $A$ | $C$ |
| $C$ | $D$ | $B$ |
| $*D$ | $D$ | $A$ |
| $E$ | $D$ | $F$ |
| $F$ | $G$ | $E$ |
| $G$ | $F$ | $G$ |
| $H$ | $G$ | $D$ |

(a) Apply the string `'00101'` to the transition profile and demonstrate $\hat{\delta}$, the extended transition function step-by-step (6 marks).

(b) Provide a detailed explanation stating whether the string in (a) above is accepted or rejected by the automaton $N$ (3 marks).

(c) Map out this configuration precisely on a structural Transition Table matrix (4 marks).

(d) Formally define the extended transition function $\hat{\delta}$ of an NFA using structural mathematical induction notation (4 marks).

(e) Explain any one major foundational architecture difference between NFAs and DFAs (3 marks).

## QUESTION TWENTY-SIX

Given the regular expression (RE): $10 + 11^*$

(a) Convert the RE to its equivalent finite state automaton engine (10 marks).

(b) Using the standard order of precedence for regular definitions, explicitly group the RE elements using parenthesization (4 marks).

(c) Using clearly stated algebraic laws for regular sets, simplify the expression (2 marks).

(d) Provide the formal English structural description of the simplified RE in (c) above (2 marks).

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Consider a context-free grammar $G$ defined with the following production laws:

$$S \to aS \mid aSbS \mid \epsilon$$

(a) Provide the complete formal 4-tuple specification of grammar $G$ (2 marks).

(b) Prove that $G$ is inherently ambiguous by showing that the string `'aab'` yields two distinct:

 i. Leftmost derivations (2 marks).

 ii. Rightmost derivations (2 marks).

## QUESTION TWENTY-SEVEN

For two regular sets $R$ and $S$, prove or disprove the following algebraic statements:

(a) $(RS + S)RS = (RRS)^*$ (5 marks)

(b) $(R + S)^* = (RRS)$ (5 marks)

For the context-free grammar defined by the core productions:

$$S \to \epsilon \mid 0 \mid 1 \mid 0S0 \mid 1S1$$

Find the explicit structural language configuration represented by this grammar (6 marks).

## QUESTION TWENTY-EIGHT

Given an alphabet $\Sigma$, formally define the following terms over $\Sigma$:

(a) Language (1 mark).

(b) Regular language (1 mark).

(c) Extended transition function of a string (2 marks).

(d) Briefly explain one implementation application of NFAs in everyday life workflows (2 marks).

(e) Differentiate fundamentally between the language sets $\{\epsilon\}$ and $\emptyset$ (2 marks).

(f) Differentiate explicitly between a mathematical relation and a mapping (2 marks).

Given two distinct operational strings $W$ = simple and $U$ = test, formed over the English alphabet, compute:

(a) $WU$ (1 mark)

(b) $W^2U$ (1 mark)

(c) $|WU|$ (1 mark)

(d) $U^T$ (1 mark)

(e) For any processing symbol $a$, prove explicitly that $(Ua)^T = a(U)^T$ (2 marks).

## QUESTION TWENTY-NINE

Given a language $L$, determine the explicit operational result of the following properties:

(a) $\emptyset \cdot L$ (2 marks)

(b) $(L^*)^*$ (2 marks)

Given two structural alphabet distributions $R = \{x, y, z\}$ and $S = \{y, z\}$, find:

(a) $(R \cup S)^*$ (2 marks)

(b) $(R \cap S)^*$ (2 marks)

(c) $(R \setminus S)^*$ (2 marks)

Consider the transition table matrix of an NFA $D$ below:

Table 11: NFA $D$ Transition Specifications

| **State** | **Input 0** | **Input 1** |
|---|---|---|
| $\rightarrow q_0$ | $\{q_0, q_3\}$ | $\{q_0, q_1\}$ |
| $q_1$ | $\emptyset$ | $\{q_2\}$ |
| $q_2$ | $\{q_2\}$ | $\{q_2\}$ |
| $q_3$ | $\{q_4\}$ | $\emptyset$ |
| $q_4$ | $\{q_4\}$ | $\{q_4\}$ |

(a) Using the state minimization algorithm framework, identify all equivalent states of $D$ (10 marks).

(b) Construct the complete minimum-state equivalent DFA system layout from your data (4 marks).

## QUESTION THIRTY

### SECTION A: Answer ALL questions in this section

Define the following foundational terms as they relate to the theory of Complexity and Computability:

(a) Running time of a Turing Machine (2 marks).

(b) Instantaneous Description of a Pushdown Automaton (PDA) (3 marks).

(c) Regular Language (1 mark).

Construct a finite automaton engine directly from the structural grammar given by these production rules:

- $Z \rightarrow U0 \mid V0$
- $U \rightarrow Z1 \mid 1$
- $V \rightarrow Z0 \mid 0$

where $\Sigma = \{0, 1\}$ and $N = \{U, V, Z\}$ (5 marks).

Differentiate clearly between the following structural pairs (2 marks each):

(a) Decidable and undecidable problems.

(b) Tractable and intractable problems.

(c) Simple and terminal Markov production strings.

(d) A relation and a mapping configuration.

## QUESTION THIRTY-ONE

If set $R = \{a, b, c, d\}$, provide complete subset relationship examples over $R$ that are:

(a) Reflexive (1 mark).

(b) Transitive (1 mark).

(c) Symmetric (1 mark).

Consider the regular expression (RE) given by: $01^* + 1$

(a) Group the RE terms by explicit operator precedence levels (2 marks).

(b) Using the structural configuration result of (a) above, draw the complete state transition diagram for the RE (4 marks).

Give and briefly explain one concrete application of each of the following computation models in daily software environments (2 marks each):

(a) Context-free grammars.

(b) Regular expressions strings.

(c) Markov parsing algorithms.

Consider the Turing Machine computational matrix model given by: $M = (\{q_0, q_1, q_2, q_3, q_4\}, \{0, 1\}, \{0, 1, X, Y, B$ where the mapping function $\delta$ is defined as: Check the operational validation and path accep-

Table 12: Turing Machine Delta Configurations

| | |
|---|---|
| $\delta(q_0, 0) = (q_1, X, R)$ | $\delta(q_0, Y) = (q_3, Y, R)$ |
| $\delta(q_1, 0) = (q_1, 0, R)$ | $\delta(q_1, 1) = (q_2, Y, L)$ |
| $\delta(q_1, Y) = (q_1, Y, R)$ | $\delta(q_2, 0) = (q_2, 0, L)$ |
| $\delta(q_2, X) = (q_0, X, R)$ | $\delta(q_2, Y) = (q_2, Y, L)$ |
| $\delta(q_3, Y) = (q_3, Y, R)$ | $\delta(q_3, B) = (q_4, B, R)$ |

tance of the input tape string `'01101'` using extended transformation processing paths (6 marks).

## QUESTION THIRTY-TWO

### SECTION B: Answer any THREE questions from this section

Consider the formal specification parameters of a Turing Machine $M$:

$$M = (\{q_0, q_1, q_2, q_3, q_4\}, \{0, 1\}, \{0, 1, X, Y, B\}, \delta, q_0, B, \{q_4\})$$

(a) What does the configuration set $\{0, 1, X, Y, B\}$ of this description define? (2 marks)

(b) Construct the physical transition diagram for machine $M$ using state circles and labeled directional arrows (8 marks).

(c) Show the step-by-step Instantaneous Descriptions (IDs) sequence for $M$ if the input tape reads the string `'01'` (6 marks).

(d) From your resulting final ID tracking in (c) above, determine conclusively if the string `'01'` is contained in $L(M)$ (2 marks).

(e) Differentiate between recursively enumerable (RE) and recursive language sets of a Turing Machine system (2 marks).

## QUESTION THIRTY-THREE

Consider a grammar, $G$, whose active productions are given by the structural rules:

- $S \to A1B$
- $A \to 0A \mid \epsilon$
- $B \to 0B \mid 1B \mid \epsilon$

(a) Write down the complete, formal 4-tuple mathematical definition of $G$ (2 marks).

(b) Identify and clearly differentiate between the different types of grammars defined within the Chomsky Hierarchy (8 marks).

(c) Show the complete leftmost and rightmost structural derivations for the terminal string `'1001'` (4 marks).

(d) Convert grammar $G$ to a Pushdown Automaton (PDA), $P$, that accepts input strings via an empty stack mechanism (6 marks).

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## QUESTION THIRTY-FOUR

Given an 8-state configuration finite engine characterized by the matrix below:

Table 13: Automaton Minimization Base Table

| **State** | **Input 0** | **Input 1** |
|---|---|---|
| $\rightarrow A$ | $B$ | $A$ |
| $B$ | $A$ | $C$ |
| $C$ | $D$ | $B$ |
| $*D$ | $D$ | $A$ |
| $E$ | $D$ | $F$ |
| $F$ | $G$ | $E$ |
| $G$ | $F$ | $G$ |
| $H$ | $G$ | $D$ |

(a) State the explicit algorithmic steps comprising the formal **State Minimization Algorithm** (5 marks).

(b) Using the algorithm framework stated in (a), isolate the completely equivalent state spaces of $D$. Clearly layout each state table partition row across successive optimization loops (10 marks).

(c) Draw the final optimized, minimal automaton diagram with all equivalent state regions cleanly consolidated (5 marks).

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## QUESTION THIRTY-FIVE

Consider an $\epsilon$-NFA architecture specified by the following data matrix:

Table 14: $\epsilon$-NFA Data Matrix for Question Thirty-Five

| **State** | $\epsilon$ | **Input a** | **Input b** | **Input c** |
|---|---|---|---|---|
| $\rightarrow p$ | $\{q, r\}$ | $\{q\}$ | $\{q\}$ | $\{r\}$ |
| $q$ | $\emptyset$ | $\{p\}$ | $\{r\}$ | $\{p, q\}$ |
| $*r$ | $\emptyset$ | $\emptyset$ | $\emptyset$ | $\emptyset$ |

(a) Construct the clear visual transition diagram layout for this $\epsilon$-NFA system (4 marks).

(b) i. Provide the strict definition for the $\epsilon$-closure property of an automaton state (2 marks).
ii. Calculate the precise $\epsilon$-closure sets for every state in this machine (3 marks).

(c) Convert this $\epsilon$-NFA profile into its equivalent operational DFA layout (11 marks).

## QUESTION THIRTY-SIX

### Markov String Processing Engines

(a) Describe the core mathematical execution sequence according to which a finite series of Markov productions $(P_1, P_2, \ldots, P_n)$ evaluates a processing target string $S$ (5 marks).

(b) Suppose a Markov algorithm machine $M$ is defined using the production configuration below, with the goal of tracking and counting substring matches within a stream:

- $P_1 : \alpha\beta\gamma \to \beta * \alpha\gamma \quad (\alpha, * \in \Sigma')$
- $P_2 : \alpha x \to x\alpha \quad (x \in \Sigma)$
- $P_3 : x \to x$
- $P_4 : \alpha \to \bullet\epsilon$
- $P_5 : \epsilon \to \alpha$

Where:

- In rule $P_1$, substring match target $\delta = \beta\gamma$, where $\beta =$ 1st character and $\gamma =$ remaining characters.
- The symbol $*$ increments the occurrence count of the match.

**Task:** Apply this procedural processing engine directly to the input sequence $S =$ '101001010101' to compute the occurrences of the target substring sequence $\delta =$ '101'. Let $\beta = 1$ and $\gamma = 01$ (15 marks).

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## QUESTION THIRTY-SEVEN

[This section explicitly mapped directly into the structural criteria specified above in Question Thirty-Six.]

## QUESTION THIRTY-EIGHT

**Instructions: Answer ALL questions (40 marks)**

Define the following terms in relation to grammars (1 mark each):

a) Syntax

b) Language

c) Derivation

d) Sentence

e) Sentential form

Given a grammar, $G$, specified as $G = (\{U, V, Z\}, \{0, 1\}, P, S)$ where production set $P$ is given by:

- $Z \to U0 \mid V0$
- $U \to Z1 \mid 1$
- $V \to Z0 \mid 0$

a) Draw the complete state transition diagram representing the finite state automaton of $G$ (5 marks).

b) Check the acceptance sequence of the input string `'0010'` and determine if it evaluates to true (2 marks).

## QUESTION THIRTY-NINE

(a) Draw a comprehensive state transition graph to represent the regular language expression $(a + b)^*bab$ (8 marks).

(b) Obtain the regular expression of the strings recognized by the transition graph below (15 marks):

Table 15: Transition Rules Table for Question Thirty-Nine

| **State** | $\epsilon$ | **Input a** | **Input b** | **Input c** |
|---|---|---|---|---|
| $\to p$ | $\emptyset$ | $\{p\}$ | $\{q\}$ | $\{r\}$ |
| $q$ | $\{p\}$ | $\{q\}$ | $\{r\}$ | $\emptyset$ |
| $*r$ | $\{q\}$ | $\{r\}$ | $\emptyset$ | $\{p\}$ |

(c) Give 4 distinct strings that are accepted by the following structural transition graph layout (2 marks).

Table 16: $\epsilon$-NFA Table Matrix for Question Forty

| State | Input 0 | Input 1 |
|---|---|---|
| $\rightarrow q_0$ | $\{q_0, q_3\}$ | $\{q_0, q_1\}$ |
| $q_1$ | $\emptyset$ | $\{q_2\}$ |
| $*q_2$ | $\{q_2\}$ | $\{q_2\}$ |
| $q_3$ | $\{q_4\}$ | $\emptyset$ |
| $*q_4$ | $\{q_4\}$ | $\{q_4\}$ |

## QUESTION FORTY

Given the operational characteristics of the $\epsilon$-NFA model below:

a. Represent this $\epsilon$-NFA engine visually as a well-formed transition diagram graph (3 marks).

b. Compute the full $\epsilon$-closure sets for each state in the system (3 marks).

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## QUESTION FORTY-ONE

Consider an NFA configuration $N$:

a. State the formal 5-tuple structural definition of $N$, explaining each item clearly (5 marks).

b. Given that $N$ has the following transition table representing its internal function:

Table 17: NFA Function Grid for Question Forty-One

| State | Input 0 | Input 1 |
|---|---|---|
| $\rightarrow q_0$ | $\{q_0, q_3\}$ | $\{q_0, q_1\}$ |
| $q_1$ | $\emptyset$ | $\{q_2\}$ |
| $*q_2$ | $\{q_2\}$ | $\{q_2\}$ |
| $q_3$ | $\{q_4\}$ | $\emptyset$ |
| $*q_4$ | $\{q_4\}$ | $\{q_4\}$ |

i) Draw the complete state transition diagram of $N$ based on these configurations (5 marks).

ii) Verify the validation and acceptance trace of the string `'01001'` using its extended transition paths.

## QUESTION FORTY-TWO

Consider a complex structural transition graph processing system $G$:

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Table 18: System Matrix Configuration for Question Forty-Two

| State | $\epsilon$ | Input a | Input b | Input c |
|---|---|---|---|---|
| $p$ | $\emptyset$ | $\{p\}$ | $\{q\}$ | $\{r\}$ |
| $q$ | $\{p\}$ | $\{q\}$ | $\{r\}$ | $\emptyset$ |
| $*r$ | $\{q\}$ | $\{r\}$ | $\emptyset$ | $\{p\}$ |

b. Using your table of derivatives obtained in (a) above, obtain the regular expression generated by the transition graph $G$ (5 marks).

c. Draw the transition graph to represent the expression/string $(0 + 11)^*1$. (5 marks)

## QUESTION FORTY-THREE

Table 19: System Matrix Configuration for Question Forty-Three

| State | $\epsilon$ | Input a | Input b | Input c |
|---|---|---|---|---|
| $p$ | $\emptyset$ | $\{p\}$ | $\{q\}$ | $\{r\}$ |
| $q$ | $\{p\}$ | $\{q\}$ | $\{r\}$ | $\emptyset$ |
| $*r$ | $\{q\}$ | $\{r\}$ | $\emptyset$ | $\{p\}$ |

a. Generate the table of derivatives to indicate the vertices of $G$ that can be reached from the initial node by following a path corresponding to the strings $x \in \Sigma$, where $\Sigma = \{0, 1\}$ (10 marks).

b. Using the table of derivatives obtained in (a) above, obtain the regular expression generated by the transition graph $G$ (5 marks).

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## QUESTION FORTY-FOUR

Given the $\epsilon$-NFA below:

Table 20: Final Configurable Rule Matrix for Question Forty-Four

| **State** | $\epsilon$ | **Input a** | **Input b** | **Input c** |
|---|---|---|---|---|
| $\rightarrow p$ | $\emptyset$ | $\{p\}$ | $\{q\}$ | $\{r\}$ |
| $q$ | $\{p\}$ | $\{q\}$ | $\{r\}$ | $\emptyset$ |
| $*r$ | $\{q\}$ | $\{r\}$ | $\emptyset$ | $\{p\}$ |

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# Part 7: Pushdown Automata & Context-Free Properties

## QUESTION FORTY-FIVE

Design Pushdown Automata (PDA) for the following languages. Clearly specify the 7-tuple definition and the transition function $\delta$ for each.

(a) $L_1 = \{a^n b^n c^m \mid n, m \geq 1\}$, accepted by empty stack (6 marks).

(b) $L_2 = \{wcw^R \mid w \in \{a, b\}^*\}$, accepted by final state (6 marks).

(c) $L_3 = \{a^i b^j c^k \mid i = j \text{ or } j = k\}$, accepted by final state (6 marks).

## QUESTION FORTY-SIX

The Pumping Lemma for Context-Free Languages (Bar-Hillel Lemma) is a crucial tool for proving non-context-freeness.

(a) State the Pumping Lemma for Context-Free Languages formally (3 marks).

(b) Use the Pumping Lemma to prove that the language $L = \{a^i b^j c^k \mid i < j < k\}$ is not a context-free language (7 marks).

(c) Explain structurally why the Pumping Lemma for CFLs requires two "pumpable" substrings ($v$ and $x$) compared to just one for regular languages (3 marks).

## QUESTION FORTY-SEVEN

Closure properties of Context-Free Languages (CFLs) differ significantly from Regular Languages.

(a) Prove that Context-Free Languages are closed under Union, Concatenation, and Kleene Star (6 marks).

(b) Provide a concrete counterexample to prove that Context-Free Languages are **not** closed under Intersection (4 marks).

(c) Using the result from (b) and De Morgan's laws, prove that CFLs are not closed under Complementation (3 marks).

## Part 8: Advanced Computability, Decidability & Complexity

### QUESTION FORTY-EIGHT

The Myhill-Nerode Theorem provides an algebraic alternative to the Pumping Lemma for proving non-regularity and determining minimal DFA states.

(a) Define the indistinguishability relation $\equiv_L$ for a language $L$ (2 marks).

(b) State the Myhill-Nerode Theorem (3 marks).

(c) Use the Myhill-Nerode theorem to prove that the language $L = \{0^n 1^n \mid n \geq 0\}$ is not regular by showing it has infinitely many equivalence classes (5 marks).

### QUESTION FORTY-NINE

Post's Correspondence Problem (PCP) is a classic undecidable problem. Consider the following PCP instance defined by the set of dominoes $D$:

$$D = \left\{ \left[\frac{1}{101}\right], \left[\frac{0}{11}\right], \left[\frac{101}{0}\right], \left[\frac{01}{10}\right] \right\}$$

(a) Formally define what constitutes a "solution" to a PCP instance (3 marks).

(b) Find a valid sequence of indices that forms a solution for the given instance $D$, or prove that no solution exists (7 marks).

### QUESTION FIFTY

Computational Complexity theory classifies problems based on their inherent difficulty and resource bounds.

(a) Formally define the complexity classes **P**, **NP**, and **NP-Complete** (6 marks).

(b) Explain the concept of "polynomial-time reducibility" ($\leq_p$) and its role in proving a problem is NP-Complete (4 marks).

(c) If an NP-Complete problem is shown to be solvable in deterministic polynomial time, what are the mathematical implications for the relationship between complexity classes P and NP? (3 marks).